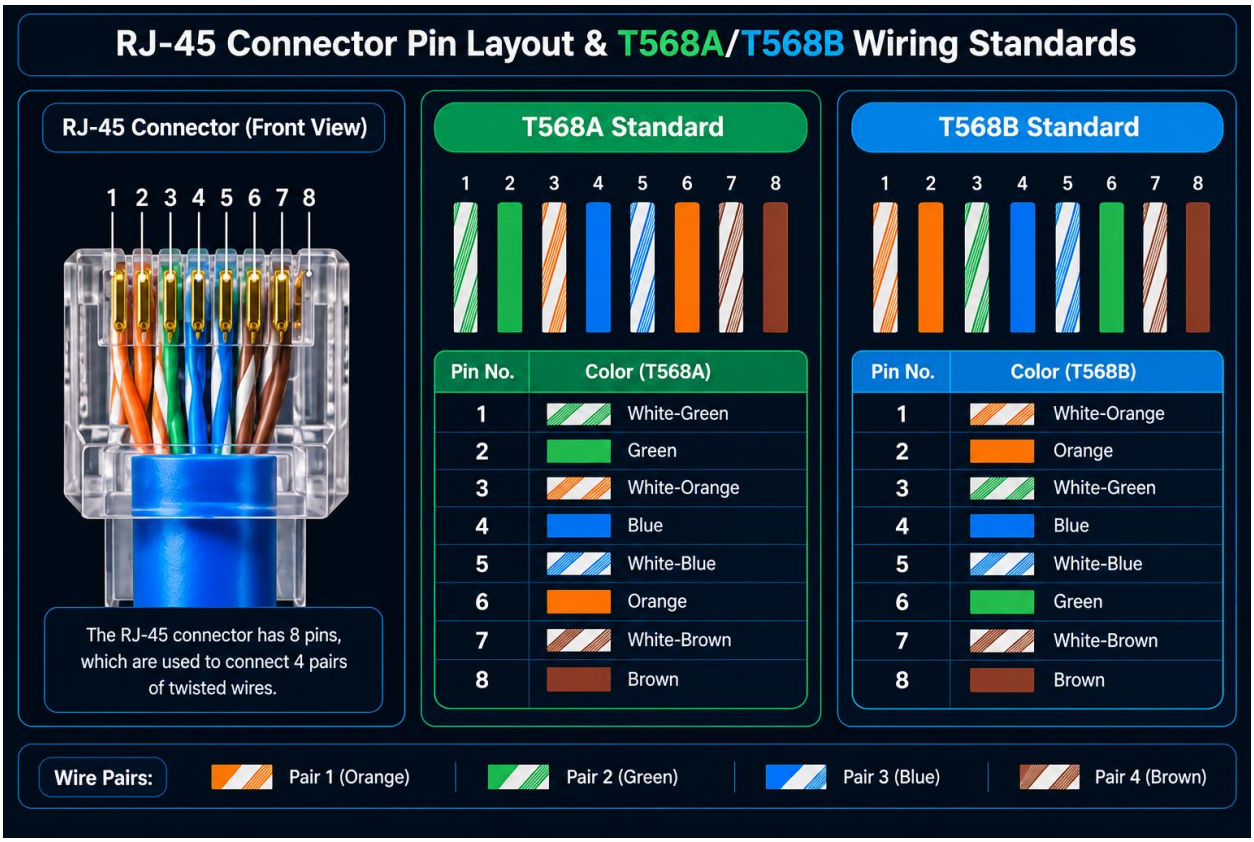
1 Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



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2 Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

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Comparison of Ethernet Standards

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable Type	Cat3 or higher UTP	Cat5 UTP	Cat5e or higher UTP
Maximum Cable Length	100 meters	100 meters	100 meters
Connector Type	RJ-45 (8P8C)	RJ-45 (8P8C)	RJ-45 (8P8C)
Common Use	Older LANs	Fast Ethernet networks	Modern LANs, offices, and gaming

3 Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.

Crossover Cable :

- **Cable Type** : Use an Ethernet crossover cable to connect two laptops directly.

- **Reason** : A crossover cable swaps the transmit (TX) and receive (RX) wire pairs, allowing one laptop's transmit signals to reach the other laptop's receive pins.

4 List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Device	Likely Ethernet Standard
Wi-Fi Router	1000BASE-T
Desktop Computer	1000BASE-T
Network Printer	100BASE-TX